产工作业

1-产品工程和传统化学工程

Based on the following reading materials, please write a short article to make a brief introduction about what is "chemical product engineering" to the readers in terms of its study objectives, scope, necessity, and differences with traditional chemical engineering. (within 300 words)

请在以下阅读材料的基础上，写一篇短文，向读者简要介绍“化学产品工程”的研究目标、范围、必要性、与传统化学工程的区别。（300 字以内）

Chemical product engineering is an approach to designing chemical products to meet specific needs and create value. 化工产品工程是设计化工产品以满足特定需求和创造价值的一种方法。It covers everything from the development of new specialty chemicals to the design of product micro-structures and the manufacture of chemical devices.它涵盖了从新型特种化学品的开发到产品微观结构的设计和化学装置的制造。

Unlike traditional chemical engineering, chemical product engineering focuses more on the design and development of products than on the optimization of chemical processes. 与传统的化学工程不同，化学产品工程更多地关注产品的设计和开发，而不是化学工艺的优化。The scope of research in chemical product engineering includes optimization of product performance, cost, and sustainability, as well as modeling of product models, process models, and sustainability metrics. 化学产品工程的研究范围包括产品性能、成本和可持续性的优化，以及产品模型、过程模型和可持续性指标的建模。

Compared to traditional chemical engineering, chemical product engineering focuses more on product performance and cost, as well as sustainability. 与传统的化工工程相比，化工产品工程更注重产品的性能和成本，以及可持续性。

The need for chemical product engineering lies in the fact that the chemical industry needs to constantly innovate and improve as market demands change.对化工产品工程的需求在于，随着市场需求的变化，化工行业需要不断的创新和改进。

Chemical product engineering provides a systematic approach to designing chemical products to meet these needs.化工产品工程为设计化工产品以满足这些需求提供了一种系统的方法。

Chemical product engineering is an emerging field of chemical engineering that utilizes computer-aided and data science techniques to design and optimize chemical products in order to improve performance, cost, and sustainability, shorten product development cycles, and reduce costs.化工产品工程是化工工程的一个新兴领域，它利用计算机辅助和数据科学技术来设计和优化化工产品，以提高性能、成本和可持续性，缩短产品开发周期，并降低成本。

Chemical product engineering is a discipline with a wide range of applications that offers new research directions and career opportunities for chemical engineers, as well as a strong impetus for the continued development of the chemical industry.化工产品工程是一门应用广泛的学科，为化学工程师提供了新的研究方向和职业机会，也为化学工业的持续发展提供了强大的动力。

Comparison Between Chemical Product Design and Chemical Process Design

Chemical product design chemical process design are both crucial aspects of chemical engineering field, but they focus on different objectives and follow distinct procedures. Here is a brief comparison:
化工产品设计化学工艺设计都是化学工程领域的关键方面，但它们关注不同的目标，遵循不同的程序。下面是一个简短的比较：

Objectives 目标

- **Chemical Product Design:**
  - Focuses on developing new products that meet specific needs and requirements of end consumers.
  - Emphasizes functionality, marketability, and consumer satisfaction.

-**化工产品设计：** -专注于开发满足终端消费者特定需求的新产品。 -强调功能、可销售性和消费者满意度。

- **Chemical Process Design:**
  - Concentrates on designing efficient processes for manufacturing chemical products.
  - Aims to optimize production efficiency, cost-effectiveness, safety, and environmental impact.

-**化学工艺设计：** -专注于设计制造化工产品的高效工艺。 -旨在优化生产效率、成本效益、安全性和环境影响。

Main Focus 主要重点

- **Chemical Product Design:**
  - Innovation in product functionality and performance.
  - Understanding user needs and translating them into product specifications.
  - Ensuring product safety, regulatory compliance, and market appeal.

-**化工产品设计：** -在产品功能和性能方面的创新。 -了解用户的需求，并将其转化为产品规格。 -确保产品安全、法规遵从性和市场吸引力。

- **Chemical Process Design:**
  - Development of process flowsheets and layouts.
  - Selection and sizing of equipment and instrumentation.
  - Process optimization for yield, purity, energy consumption, and waste management.

-**化学工艺设计：** -制定工艺流程和布局。 -设备和仪器仪表的选择和尺寸调整。 -工艺优化的产量、纯度、能耗和废物管理。

Design Procedures 设计程序

- **Chemical Product Design:**
  1. **Market Research**: Identifying market needs and opportunities.
  2. **Concept Development**: Generating ideas and concepts for new products.
  3. **Feasibility Study**: Evaluating technical and economic feasibility.
  4. **Design and Prototyping**: Creating detailed designs and prototypes.
  5. **Testing and Validation**: Conducting tests to ensure product meets specifications.
  6. **Commercialization**: Launching the product into the market.

化学品设计： 1.**市场研究**：确定市场需求和机会。 2.**概念开发**：为新产品产生想法和概念。 3.**可行性研究**：评估技术和经济上的可行性。 4.**设计和原型设计**：创建详细的设计和原型。 5.**测试和验证**：进行测试，以确保产品符合规格要求。 6.**商业化**：将产品投放市场。

- **Chemical Process Design: -**化学工艺设计：****

  1. **Process Synthesis**: Developing preliminary process flowsheets.
  2. **Process Simulation**: Using software to model and simulate processes.
  3. **Equipment Selection**: Choosing appropriate equipment and materials.
  4. **Process Optimization**: Refining processes to enhance efficiency and reduce costs.
  5. **Safety and Environmental Assessment**: Ensuring compliance with safety and environmental regulations.

6. **Detailed Engineering**: Producing detailed engineering drawings and specifications.

1.**工艺合成**：开发初步工艺流程图。 2.**过程模拟**：使用软件对过程进行建模和模拟。 3.**设备选择**：选择合适的设备和材料。 4.**工艺优化**：精炼工艺，以提高效率和降低成本。 5.**安全和环境评估**：确保符合安全和环境法规。 6.**详细工程**：制作详细的工程图纸和规范。

Interrelationship####相互关系

- The success of chemical product design often depends on the efficiency and feasibility of the chemical process design.-化学产品设计的成功往往取决于化学工艺设计的效率和可行性。
- Both fields need to work together closely; a well-designed product requires an effective and sustainable manufacturing process. -这两个领域需要紧密合作；一个设计良好的产品需要一个有效和可持续的制造过程。
- Innovations in product design can drive new requirements for process design, and advances in process technologies can enable new product possibilities. -产品设计方面的创新可以推动工艺设计的新需求，而工艺技术方面的进步可以使新的产品成为可能。
- Coordination between these two areas ensures that new products can be manufactured reliably, safely, and cost-effectively while meeting market demands. -这两个领域之间的协调确保新产品能够在满足市场需求的同时可靠、安全、经济有效地满足市场需求。

In summary, chemical product design is oriented towards creating new products that fulfill consumer needs, whereas chemical process design focuses on developing processes for producing these products efficiently. Both disciplines are interconnected and rely on each other for successful implementation and commercialization of new chemical products.

总之，化学产品设计的方向是创造满足消费者需求的新产品，而化学工艺设计的方向是专注于开发高效生产这些产品的工艺。这两个学科是相互关联的，并相互依赖于新的化学产品的成功实施和商业化

作业题 2

2.6 Degumming Soybean Oil.脱脂大豆油。

According to the literature "Application of Membrane Separation Technology in Edible Oil Degumming and Solvent Recovery," revealed that the maximum retention of phospholipids in a soybean oil-hexane blend, achieved through ultrafiltration without water addition, was 63.5%. Assuming a 10,000-ton soybean oil-hexane mixture with a 40% phospholipid content, the initial ultrafiltration yield would be 7,460 tons. Furthermore, according to 2021 statistics, China produced 17.938 million tons of soybean oil. Integrating these findings, it is evident that nanofiltration purification of the ultrafiltered soybean oil extract results in an 81% retention rate. Consequently, it can be estimated that the preliminary production of ultrafiltered soybean oil extract totals 22.146 million tons.

根据文献《膜分离技术在食用油脱胶和溶剂回收中的应用》，大豆油-己烷混合物-超滤中磷脂的最大保留率为 63.5%。假设 10000 吨大豆油-己烷混合物的磷脂含量为 40%，初始超滤产率将为 7460 吨。此外，根据 2021 年统计，中国大豆油 1793.8 万吨。综合这些发现，很明显，纳滤纯化超滤大豆油提取物的 81%的保留率。因此，可以估计，超滤大豆油提取物的初步产量总计为 2214.6 万吨。

2.7 Biodegradable Plastics.

The major markets for biodegradable plastics from renewable feedstocks are likely to include:可再生原料中的生物可降解塑料的主要市场可能包括：

1. Industry 工业 食品包装：袋子，包装膜，容器。 非食品包装：电子产品、商品包装、邮寄材料。
Food packaging: Bags, wrapping films, containers.
Non-food packaging: Electronics, goods wrapping, mailing materials.
2. Agricultural Sector 农业部门 覆盖膜：用于农业以改善土壤条件的可生物降解膜。控释农化载体。
Mulch films: Biodegradable films used in agriculture to enhance soil conditions.
Controlled-release agrochemical carriers.
3. Consumer Goods 消费品 一次性餐具和盘子。 可压缩垃圾袋。
Disposable cutlery and plates.
Compostable garbage bags.
4. Medical Industry 医疗行业 生物可吸收的医疗植入物和缝合线。 一次性医疗服、手套和口罩。
Bioresorbable medical implants and sutures.
Disposable medical gowns, gloves, and masks.
5. Textile Industry 纺织工业 环保服装用的纤维。 卫生用品用无纺布纺织品。
Fibers for eco-friendly clothing.
Non-woven textiles for hygiene products.

Major needs that must be satisfied for biodegradable plastics to succeed include:可生物降解塑料的成功必须满足的主要需求包括：

1. Availability: Sufficient production capacity to meet demand.可用性：有足够的生产能力来满足需求。
2. Regulatory Approval: Compliance with international standards and obtaining certifications to assure customers of the product's environmental claims.监管批准：遵守国际标准并获得认证，向客户确保产品的环境要求。
3. Cost Competitiveness: The price point must be competitive with conventional plastics to encourage widespread adoption.成本竞争力：价格点必须与传统塑料具有竞争力，以鼓励广泛采用。
4. Performance: The materials need to provide similar or superior properties compared to traditional plastics like strength, flexibility, and shelf life.性能：这些材料需要提供与传统塑料类似或更优越的性能，如强度、柔韧性和保质期。
5. Compostability: The biodegradation process should occur within a reasonable time frame and under appropriate conditions such as industrial composting facilities.堆肥稳定性：生物降解过程应在合理的时间框架内和适当的条件下进行，如工业堆肥设施。

3.8 Clean Feet（为治疗脚臭提供 idea）

1. Wash feet daily: Use an antibacterial soap to reduce bacteria which can cause the smell.

2. Dry feet properly: Moisture encourages bacterial growth. Make sure to pay attention to the area between your toes.

3. Change socks regularly: Fresh socks help absorb sweat and reduce odor.

4. Use talcum powder: This helps keep your feet dry and reduces odor.

5. Exfoliate your feet: Regularly remove dead skin cells that bacteria feed on.

6. Use foot deodorizing spray: These products kill bacteria and provide a fresh scent.

7. Soak your feet in Epsom salt: This can reduce sweating and smells.

8. Try antifungal and antibacterial creams: These can control any infections causing the odor.

9. Use odor-absorbing insoles: These products capture and neutralize foot odors.

10. Rotate shoes: Give each pair of shoes at least 24 hours to air out before wearing them again.

11. Stay hydrated: Proper hydration can reduce sweating and its associated odors.

12. Consult a podiatrist: Get professional advice for persistent foot odor.

13. Use prescription antiperspirants: These can control excessive sweating.

14. Take oral medications: Certain antibiotics or other drugs may be recommended by a doctor.

15. Try apple cider vinegar: Soak your feet in it to balance the pH and reduce odor.

16. Wear breathable shoes: Shoes made of materials like leather or canvas can reduce sweat and odor.

17. Choose moisture-wicking socks: They help keep your feet dry.

18. Try odor-absorbing shoe sprays: Spray inside your shoes to combat smells.

19. Use foot powders: These absorb moisture and reduce odor.

1.每天洗脚：使用抗菌肥皂来减少可能引起气味的细菌。 2.正确干燥的脚：水分促进细菌生长。一定要注意你的脚趾之间的区域。 3.定期更换袜子：新鲜的袜子有助于吸收汗水，减少异味。 4.使用滑石粉：这有助于保持你的脚的干燥，减少气味。 5.剥离你的脚：定期去除被细菌吞噬的死皮细胞。 6.使用脚部除臭喷雾剂：这些产品可以杀死细菌，并提供一种新鲜的气味。 7.将脚浸泡在泻盐中：这样可以减少出汗和气味。 8.尝试抗真菌和抗菌霜：这些可以控制任何引起气味的感染。 9.使用吸臭鞋垫：这些产品可以捕捉和中和脚部的气味。 10.旋转鞋：每双鞋至少 24 小时，再次透气。 11.保持水分：适当的水分可以减少出汗及其相关的气味。 12.请咨询足病医生：就持续的足部气味获得专业的建议。 13.使用处方止汗剂：这些药物可以控制过度出汗。 14.服用口服药物：可建议使用某些抗生素或其他药物

3.9 Insulin Aerosols 胰岛素气溶胶（方案筛选，筛选矩阵）

The above programs can be categorized as follows:

1.Oral Administration:1.口腔用药： 想法 2：将胰岛素附着在另一个分子上以供口服。 想法 14：将胰岛素封装在囊泡中进行口服摄入。

Idea 2: Attach insulin to another molecule for oral ingestion.

Idea 14: Encapsulate insulin in vesicles for oral intake.

2. Transdermal Systems: 2.透皮系统： 想法 8：创建一个非常小的植入式泵，通过皮肤注射胰岛素。想法 9：采用电穿孔法推动胰岛素穿过皮肤。 想法 10：修改胰岛素，使其能够通过皮肤扩散。 想法 11：开发一种类似于尼古丁贴片的胰岛素贴片。

Idea 8: Create a very small implantable pump that administers insulin through the skin.

Idea 9: Employ electroporation to push insulin across the skin.

Idea 10: Modify insulin to enable diffusion through the skin.

Idea 11: Develop an insulin patch similar to a nicotine patch.

3. Inhalation Systems:3.吸入系统： 想法 12：将胰岛素作为气溶剂吸入鼻腔。 4.修改后的针头方法： 想法 18：使用许多非常小的针头。 想法 19：设计无针注射机构。

Idea 12: Deliver insulin as an aerosol for nasal inhalation.

4. Modified Needle Approaches:4.修改后的针头方法： 想法 18：使用许多非常小的针头。 想法 19：设计无针注射机构。

Idea 18: Utilize many very small needles.

Idea 19: Design needle-free injection mechanisms.

5. Implantable Devices:5.植入式设备： 想法 5：建立一个通过手术植入的泵，可以按需释放胰岛素。 想法 6：植入物修饰的胰岛素对葡萄糖水平敏感，以便及时溶解。 想法 13：构建一个可以通过手术植入来发射胰岛素脉冲的设备。 想法 20：用一个病人可以手动控制的瓣膜制造一个胰岛素储存器。

Idea 5: Build a surgically implanted pump that releases insulin on-demand.

Idea 6: Implant modified insulin sensitive to glucose levels for timely dissolution.

Idea 13: Construct a device that can be surgically implanted to emit insulin pulses.

Idea 20: Make an insulin reservoir with a valve that the patient can control manually.

6. Controlled Release Systems:6.受控释放系统： 想法 4：制定缓释胰岛素，以减少注射的频率。 想法 17：注入微球，慢慢释放胰岛素进入系统。

Idea 4: Formulate slow-release insulin to reduce the frequency of injections.

Idea 17: Inject microspheres that slowly release insulin into the system.

7. Genetic Engineering:7.基因工程： 想法 3：利用基因工程来解决胰岛素的生产或行政管理问题。

Idea 3: Utilize genetic engineering to address insulin production or administration issues.

8. Artificial Pancreas Development:8.人工胰腺发育： 想法 15：开发一个完全可以自动控制血糖水平的人工胰腺。

Idea 15: Develop a fully artificial pancreas that manages blood glucose levels autonomously.

9. Alternative Drug Therapies:替代药物治疗： 想法 7：探索使用其他药物作为胰岛素的替代品或提高其疗效。 想法 16：使用不同药物的组合可能会提高治疗效果。

Idea 7: Explore the use of other drugs as alternatives to insulin or to enhance its efficacy.

Idea 16: Use a combination of different drugs possibly to improve treatment efficacy.

10. Needleless Options:10.无针选项： 想法 1：不使用传统针头注射胰岛素（无针系统）。

Idea 1: Inject insulin without using traditional needles (needle-free systems).

As we know, insulin cannot be absorbed by the body through the small intestine, thus oral administration seems to be impossible. 我们知道，胰岛素不能通过小肠被人体吸收，因此口服似乎是不可能的。Besides, artificial pancreas development, implantable devices and genetic engineering are too expensive for most people with chronic diseases to afford.此外，人工胰腺的开发、可植入的设备和基因工程对大多数慢性病患者来说过于昂贵，因此负担不起。According to the daily use needs of most diabetic patients

and the family economic situation, our task transdermal system development is the best choice.根据大多数糖尿病患者的日常使用需求和家庭经济状况，我们的任务透皮系统开发是最佳选择。The following table is a summary of feedback from diabetic patients and expert advice.下表是来自糖尿病患者的反馈和专家建议的总结。

criterion 准则	weights 权重	Option 1	Option 2	Option 3	Option 3
healthy	0.5	8	5	8	8
effect	0.4	8	9	9.5	9.5
expenses	0.1	4	4	8	9
totals	—	7.6	6.5	8.6	8.7

In summary, for the above three options, developing an insulin patch similar is a more desirable option to ensure that its treatment efficiency is not too different from injections, while reducing the scars left by the patient's treatment, as well as ensuring the health of the patient's use and reducing the associated expenses.

总之，对于上述三个选项，开发一个胰岛素贴片类似是一个更可取的选择，以确保其治疗效率不是太不同于注射，同时减少病人留下的伤疤的治疗，以及确保病人的健康使用和减少相关的费用。

3.10 Wine Aeration 葡萄酒曝气（方案分类）

Here's an organization of the ideas:

The reason for inflating wine:

1. Polyphenols react with oxygen to improve flavor.
2. The kinetics of flavor are faster than making vinegar.
3. Aerated wine tastes “round”.

Type of gas filled:

1. Want to add oxygen.
2. Want some entrained air (not too much).
3. Aerate with N2, not O2.
4. Aerate with CO2.

The basis for selecting gases:

1. Better taste means less acid.

Method of filling gas:

1. Better if poured three times.
2. Use a “ship’s decanter”.
3. Use a compact falling film evaporator.
4. Put the wine in a blender.
5. We need an oxygen sensor.
6. Use a fish bowl aerator.
7. Use structured packing for gas absorption.
8. Add a pill containing iron filings.
9. Use selective adsorption.
10. Add oxygen with a syringe.
11. Add oxygen with a straw.
12. Add a pill that gives off oxygen.
13. Add bamboo leaves (they work in green tea).
14. Use a selective absorbent.
15. Stir the wine magnetically,so we get a vortex.
16. Use hollow-fiber membranes.
17. Oxygen should come from the cork, which could be a special plastic.

以下是一个组织的组织： 葡萄酒膨胀的原因： 多酚与氧气发生反应以改善风味。 味道的动力学比做醋要快。 充气葡萄酒尝起来“圆”。 填充气体类型： 想要添加氧气。 想要一些携带的空气（不是太多）。 用 n2 曝气，而不是 o2。 用二氧化碳充气。 选择气体的依据： 味道好一点意味着少一点酸。 充气方式： 如果倒三次就会更好。 使用“船的甲板船”。 使用一个紧凑的下降膜蒸发器。 把酒放在搅拌机里。 我们需要一个氧气传感器。 使用鱼缸曝气器。 采用结构化填料进行气体吸收。 加入一个含有铁薄片的药片。 使用选择性吸附。 用注射器加入氧气。 用吸管加入氧气。 加入一粒能释放氧气的药丸。 加入竹叶（它们在绿茶中有效）。 使用选择性吸收剂。 用磁性的方法搅拌葡萄酒，这样我们就会产生一个漩涡。 使用中空纤维膜。 氧气应该来自软木塞，这可能是一种特殊的塑料。

Action or effect:

1. This only works for red wine.
2. Same effect works for green tea.
3. Pouring gives better flavors but less bouquet.
4. Will it work better if hot?

Gas acquisition method:

1. Use electrolysis to make oxygen.

Further issues to consider:

1. It must depend on the kind of red wine.
2. Does it depend on the size of bottles?
3. Is taste or smell more important?
4. Is the market for making cheap or expensive wine better?
5. Do winos have the same preferences as gourmets?
6. In China, wine is served hot.
7. In Japan, sake is served hot.
8. The machine for wine breathing should be in the wine store.
9. No, it should be at home; that’s a bigger market.这只适用于红酒。 同样的效果也适用于绿茶。 浇酒味道更好，但香味更少。 如果热的话会更好吗？ 气体采集方法： 用电解法制造氧气。 需要考虑的进一步问题： 这一定取决于红酒的种类。 这取决于瓶子的大小吗？ 味道或气味更重要吗？ 生产便宜或昂贵的葡萄酒的市场更好吗？ 葡萄酒和美食家有同样的偏好吗？ 在中国，葡萄酒是热饮用的。 在日本，清酒是热的。 葡萄酒呼吸机应该在葡萄酒商店里。 不，它应该是在家里；这是一个更大的市场。

Because of the need to be partially oxidized in red wine, we cannot choose attempts to introduce reducing agents or replace oxygen with non-oxidizing gases. Adding a pill that gives off oxygen may be a more appropriate choice. By establishing a relationship with the content of oxidation products corresponding to the best taste as the target function and the quality of the oxidized content contained in a single pill as the independent variable, we can keep the taste of each batch of red wine relatively stable.由于在红酒中需要部分氧化，我们不能选择尝试引入还原剂或用非氧化气体代替氧气。添加一种能释放氧气的药丸可能是一个更合适的选择。通过建立与最佳口感对应的氧化产物含量为目标函数，

以单丸中氧化含量的质量为自变量的关系，可以保持每批红酒的口感相对稳定。

4.3

There are two ways to produce innoxious blood bag raw materials. One is to replace di(2-ethylhexyl) phthalate with a non-carcinogenic plasticizer; the other is to develop an alternative for polyvinyl chloride. According to the existing studies, we propose the following scheme.有两种方法可以生产无害的血袋原料。一种是用非致癌增塑剂取代邻苯二甲酸二（2-乙基己酯）；另一种是开发聚氯乙烯的替代品。根据现有的研究，我们提出以下方案。

1. replace di(2-ethylhexyl) phthalate with a non-carcinogenic plasticizer
   (1) Acetyl tributyl citrate (ATBC), as a new non-toxic and environmentally friendly plasticizer, has been recognized by most countries
   (2) Aliphatic diotate esters, such as dibutyl decanoate (DBS)
   (3) Some of the epoxides, such as dibutyl phthalate (EPS), have been shown to be used in blood bags

用非致癌增塑剂取代邻苯二甲酸二（2-乙基己酯） 柠檬酸乙酰三丁酯（ATBC）作为一种新型的无毒、环保的增塑剂，已得到大多数国家的认可 脂肪族二位酯，如癸酸二丁酯（DBS） 一些环氧化合物，如邻苯二甲酸二丁酯（EPS），已被证明可用于血袋

2. develop an alternative for polyvinyl chloride
   (1) PLA has been widely used in the field of medical devices
   (2) Some polyester materials are non-toxic and soft
   (3) The film made of non-carcinogenic material can be attached to the inside of the original blood bag

开发一种聚氯乙烯的替代品 PLA 在医疗器械领域得到了广泛的应用 一些聚酯材料无毒且柔软 由非致癌材料制成的薄膜可以附着在原血袋的内部

According to the material safety, manufacturing cost and other factors, we set up the following selection matrix：材料安全、制造成本等因素，我们建立了以下选择矩阵：

Selecting ATBC

Parameter	Weighting Factor	new plasticizer			new bag materials		
		ATBC	DBS	EPS	PLA	polyester	film
Security	0.5	8	8	8	8	8	6
Cost	0.3	6	5	5	4	3	4
Degradablility	0.2	5	5	5	7	8	5
total	1	6.8	6.5	6.5	6.6	6.5	5.2

In conclusion, we chose to use ATBC as a plasticizer to replace di(2-ethylhexyl) phthalate .

综上所述，我们选择使用 ATBC 作为增塑剂来取代邻苯二甲酸二（2-乙基己酯）。

4.8

Thermal degradation Is a key factor affecting the flavor of orange juice, and its loss at a certain temperature and oxygen can be calibrated by analytical chemistry. So this factor will serve as the main criterion. We give it a weight of 0.3.热降解是影响橙汁风味的关键因素，其在一定温度和氧气条件下的损失可以通过分析化学方法进行校准。所以这个因素将成为主要的标准。我们给它 0.3 的权重。

Volatile loss It is more obvious in the vacuum distillation process, so it can be used as the second evaluation criterion. We give it a weight of 0.25.挥发性损失在真空蒸馏过程中更为明显，可作为第二评价标准。我们给它 0.25 的权重。

Engineering difficulty Deides the feasibility of the project. We also give it a weight of 0.25.工程难度决定了该项目的可行性。我们也给出了它 0.25 的权重。

Product cost It will affect the market competitiveness of the final product, but considering that the concentrated orange juice made with the new separation technology may be loved by more consumers, and it does not need to add fresh orange juice to get a better taste, because we only give it a weight of 0.2 产品成本将影响最终产品的市场竞争力，更多的消费者可能会喜欢但考虑到浓缩橙汁与新的分离技术，它不需要添加新鲜橙汁得到一个更好的味道，因为我们只给它一个 0.2 的重量

Selecting Reverse Osmosis

Parameter	Weighting Factor	distillation	freeze drying	reverse osmosis
Thermal degradation	0.3	5	8	9
Volatile loss	0.25	5	9	4
Engineering difficulty	0.25	5	5	6
Product cost	0.2	5	3	4
total	1	5	6.5	6

Following the above selection matrix, we chose freeze drying as the technique to concentrate the orange juice 根据上述选择矩阵，我们选择冷冻干燥作为浓缩橙汁的技术

5.5

(1)

On Amazon, Wal-Mart and other supermarket shopping websites, we found that the average baby bottle capacity is about 150-200ml. Based on a total surface area of 200cm^2, the modified bottle should heat the milk as quickly as possible and remain at 40 ℃. The heat capacity of supersaturated sodium acetate trihydrate is 1.972kJ/kg.在亚马逊、沃尔玛等超市的购物网站上，我们发现婴儿奶瓶的平均容量约为 150-200 毫升。根据总表面积为 200 厘米 2，改良后的瓶子应尽快加热牛奶，并保持在 40℃。过饱和三水合乙酸钠的热容为 1.972kJ/kg。

Bottle heating is an unsteady heat transfer process, assuming that we can experimentally obtain the base number Bi and Fourier number Fo of a bottle containing 80% milk.According to the following formula.瓶加热是一个不稳定的传热过程，假设我们可以实验得到含有 80%牛奶的瓶子的碱数 Bi 和傅里叶数 Fo。根据以下公式计算。

$$(t - t0)/(t^* - t0) = \exp(-BiFo)$$

Upon reaching near stabilization, the liquid temperature is approximately 40℃. And the melting point of the heating medium used is 57℃.Assuming that the insulation is required for 30 minutes, we estimated the required supersaturated sodium acetate trihydrate as 100g.当达到接近稳定时，液体温度约为 40℃。所用加热介质的熔点为 57℃。假设绝缘需要 30 分钟，我们估计所需的过饱和三水合醋酸钠为 100g。

In conclusion, the required specification of the milk bottle is about 300ml.It is equivalent to increasing the thickness of 5mm-8mm around the original bottle.The increase in bottle thickness is convenient enough for parents carrying the bottle.总之，奶瓶的要求规格约为 300 毫升。这相当于增加原瓶周围 5mm-8mm 的厚度。增加瓶子的厚度，足以方便父母携带瓶子。

(2)

We consider the risks of using this way for bottle heating as follows：

1. The filling layer may leak, leading to poor insulation
2. The cost of milk bottles is rising, resulting in a decrease in potential customers

我们认为使用这种方式进行瓶子加热的风险如下： 填充层可能会泄漏，导致绝缘性差 奶瓶的成本正在上升，导致了潜在客户的减少

Warm baby bottle (30 points). Parents of babies often wish to heat milk at around 40℃ while travelling or while in locations where there is no easily available sources of hot water except bottled water. We propose making a device that consists of a sleeve filled with supersaturated sodium acetate tri-hydrate (CH3CO2Na.3H2O). This material crystalizes at 57℃, but spontaneous nucleation does not occur above -25℃. The product is therefore stable when molten until crystallization is initiated. This can be done by bending a crack metal disk held in the solution. Once nucleated, heat is generated and the milk warmed. Please
(1) Find the true needs and the key specifications of the product;
(2) List the missing information necessary for the product design?
(3) What is the trigger of the product?
(4) What is the demerits and limitation of this product?
(5) What is the potential substitutes or competitive product of this product?
温暖的婴儿奶瓶(30 分）。婴儿的父母经常希望在周围加热牛奶 40℃在旅行或在没有容易获得的来源的地方 除了瓶装水。我们建议制造一个由 a 套筒填充过饱和三水合醋酸钠(乙酸钠。3H2O). 这种物质在 57℃处结晶，但在上面没有发生自发成核 -25℃.因此，当熔融到结晶开始时，产物是稳定的。 这可以通过弯曲溶液中固定的裂纹金属盘来实现。一旦成核， 这样就会产生热量，牛奶就会被加热。请 (1)找到产品的真实需求和关键规格； (2)列出产品设计所需的缺失信息？ (3)该产品的触发因素是什么？ (4)本产品的缺点和限制是什么？ (5)本产品的潜在替代品或竞争产品是什么？

1. True Needs and Key Specifications of the Product

1.产品的真实需求和关键规格

True Needs:

- **Convenient milk warming:** Parents need a simple, portable solution to warm milk for babies when they are traveling or in locations without easy access to hot water.-**方便的牛奶加热：**父母需要一个简单的，便携的解决方案，当婴儿旅行或在没有容易的地方获得热水。
- **Safety:** The device should be safe to use, avoiding any risk of burning or overheating the milk.-**安全：**该装置应安全使用，避免任何燃烧或过热的风险。
- **Ease of use:** It should be easy to operate with minimal steps so that it can be used quickly and efficiently.-**易用性：**它应该易于操作与最小的步骤，以便它可以快速和有效地使用。

Key Specifications:

- **Heating mechanism:** The sleeve must heat the milk to around 40℃.-**加热机构：**套筒必须将牛奶加热到 40℃左右。-

Activation method: The product should have an easy-to-operate activation mechanism (e.g., bending a metal disk).**激活方法：**该产品应具有易于操作的激活机制（如，弯曲金属圆盘）。

- **Temperature stability:** The device should maintain the milk at a stable temperature of around 40℃ once heated.-**温度稳定性：**加热后保持在 40℃左右的稳定温度。
- **Portability:** The device should be compact and lightweight for ease of carrying.**该设备应小巧轻便，便于携带。
- **Reusable:** The sodium acetate tri-hydrate solution should be reusable after initial crystallization by re-melting it.*可重复使用：**三水乙酸钠溶液在初始结晶后应可重复使用。

2. Missing Information Necessary for the Product Design2.缺少产品设计所需的信息

- **Material Details:** Specific information on the materials for the sleeve and the container to ensure food safety and heat retention.**材料细节：**关于套筒和容器材料的具体信息，以确保食品安全和保温。
- **Crystallization Control:** Detailed mechanisms to control the crystallization process accurately, ensuring consistent heating.*结晶控制：**精确控制结晶过程，确保持续加热。

- **Heat Retention Time:** Data on how long the sleeve can maintain the milk at 40℃ after heating.-**保热时间：**加热后 40℃的数据。
- **Reusability Cycles:** Information on how many times the sodium acetate tri-hydrate solution can be reused before losing efficacy.**可重复使用的循环：**关于三水合物醋酸钠溶液在失去疗效之前可以重复使用的次数的信息。
- **Safety Protocols:** Guidelines and testing results ensuring that the device does not overheat or pose any safety risks.-**安全协议：**指南和测试结果，确保设备不会过热或构成任何安全风险。

3. Trigger of the Product### 3.本产品的触发器
The trigger of the product is the **bending of a metal disk** held within the sodium acetate tri-hydrate solution. This action initiates the crystallization process, releasing heat to warm the milk.
产品的触发器是弯曲固定在三水醋酸钠溶液中的金属盘。这一作用启动了结晶过程，释放热量来加热牛奶。

4. Demerits and Limitations of this Product### 4.本产品的缺陷和局限性
- **Limited Heat Duration:** The product may not keep the milk warm for extended periods, requiring reheating if needed for longer durations.
- **Reactivation Requirement:** After each use, the sodium acetate solution needs to be re-melted, which could be inconvenient when traveling without access to boiling water.
- **Potential Leakage:** There is a possibility of leakage of the sodium acetate solution if the sleeve is damaged.
- **Initial Cost:** The product might have a higher initial cost compared to simpler, non-reusable alternatives.
-**有限加热时间：**产品可能不能长时间保温，如果需要较长时间，需要重新加热。 -**再活化要求：**每次使用后，醋酸钠溶液需要再融化，在没有沸水的旅行时可能不方便。 -**潜在泄漏：**如果套筒损坏，醋酸钠溶液有可能泄漏。 -**初始成本：**与更简单、不可重用的替代品相比，该产品可能具有更高的初始成本。

5. Potential Substitutes or Competitive Products### 5.潜在的替代品或竞争产品
- **Battery-operated bottle warmers:** These devices use battery power to heat the milk and can provide a longer-lasting heating solution.
- **Portable thermos flasks:** High-quality thermos flasks can keep pre-heated milk warm for several hours, offering a passive warming solution.
- **Chemical heat packs:** Similar products using different chemical reactions (e.g., iron oxidation in hand warmers) could offer competition.
- **Microwaveable bottles:** Bottles designed to be heated in a microwave might offer convenience where such facilities are available.
-**电池驱动的暖瓶器：**这些设备使用电池电源来加热牛奶，并可以提供一个更持久的加热解决方案。 -**便携式保温瓶：**高质量的保温瓶可以保持预热的牛奶加热数小时，提供被动加热的解决方案。 -**化学热包：**使用不同化学反应的类似产品（例如，手加热器中的铁氧化）可以提供竞争。 -**微波瓶：**设计用微波炉加热的瓶，可以在有这些设施的地方提供便利。

COVID-19 is a frustrating epidemic, which has lasted for three years worldwide and led over millions of death tolls. The global scientists have been working hard to find effective ways to cure the disease or prevent its spreading. Now, as a new graduate, you are encouraged to initiate a new company in accordance with the national calls for “mass entrepreneurship and innovation” under the help of interest free-loan initiative. As the head of the company, you are planning to build up a new product development team to develop a chemical product that includes but not limited to a new Chinese drug, a chemical medicine, healthcare wares or special devices based on your expertise/experience, education background, interests, or market needs.

1) Please list the main work to be conducted as a leader of the product team.
2) Please list the main procedure and the objectives of each step.
3) Please determine the true needs of your new product, and its key specifications.
4) Draw a diagram sketch of your product to illustrate its working principle.
5) What kinds of missing information should be collected further for the design of the product?
6) Draw a Gantt-chart for the development of the new product including the investment plan.

新冠肺炎是一种令人沮丧的流行病，在全球范围内已经持续了三年并导致了数百万人的死亡人数。全球的科学家们一直在努力工作寻找治愈疾病或防止疾病传播的有效方法。现在，作为一个新的毕业后，我们鼓励您成立一家新公司，按照规定全国呼吁“大众创业和创新”的帮助下无息贷款计划。作为公司的负责人，你正在计划进行建设组建一个新的产品开发团队来开发一种化学产品，其中包括但不限一种中国新药、一种化学药品、医疗保健产品或根据您的专业知识/经验、教育背景、兴趣、或市场需求。

1)请列出作为产品团队的领导所要进行的主要工作。
2)请列出每个步骤的主要步骤和目标。
3)请确定您的新产品的真实需求及其关键规格。
4)绘制产品的示图以说明其工作原理。
5)在设计过程中需要进一步收集哪些类型的缺失信息产品？
6) 为新的甘特图绘制了一个甘特图

1) Main Work as the Leader of the Product Team

As the leader of the product development team, the main tasks include:#### 1)作为产品团队领导的主要工作 作为产品开发团队的领导者，主要任务包括：

- **Conceptualization and Brainstorming**: Generate and refine ideas for potential chemical products.**概念化和头脑风暴**：为潜在的化学产品产生和完善的想法。
- **Research and Feasibility Study**: Conduct preliminary research to evaluate the viability of the product.-**研究和可行性研究**：进行初步研究，评估产品的可行性。
- **Team Formation**: Assemble a multidisciplinary team including chemists, biologists, engineers, and market analysts.-**团队组成**：组建一个多学科团队，包括化学家、生物学家、工程师和市场分析师。
- **Project Planning**: Develop a project plan outlining milestones, timelines, and resource allocation.-**项目计划**：制定一个项目计划，概述里程碑、时间表和资源分配。
- **Regulatory Compliance**: Ensure that the product complies with regulatory requirements.-**法规遵从性**：确保产品符合法规要求。
- **Funding and Investment**: Secure funding through interest-free loans and other sources.-**融资和投资**：通过无息贷款和其他来源获得资金。
- **Prototype Development**: Design and test prototypes of the product.-**原型开发**：设计和测试产品的原型。
- **Market Analysis**: Assess the market demand and potential competition.-**市场分析**：评估市场需求和潜在竞争。

- **Risk Management**: Identify and mitigate potential risks.-**风险管理**：识别并减轻潜在的风险。
- **Product Launch**: Plan and execute the product launch.-**产品发布**：计划并执行产品发布。

2) Main Procedure and Objectives of Each Step#### 2)各步骤的主要步骤和目标

1. **Idea Generation and Screening**
   - **Objective**: Generate innovative ideas for chemical products and screen them based on feasibility and impact.1.**创意的生成和筛选 目标**：为化工产品的创新思路，并根据可行性和影响进行筛选。
2. **Feasibility Study**
   - **Objective**: Conduct technical and market feasibility studies to determine the viability of the shortlisted ideas.进行技术和市场可行性研究，以确定入围创意的可行性。
3. **Team Formation**建立一个具有所需专业知识的称职团队来开发产品。
   - **Objective**: Build a competent team with the required expertise to develop the product.
4. **Project Planning**定义项目范围、时间轴、里程碑和资源分配。
   - **Objective**: Define the project scope, timeline, milestones, and resource allocation.
5. **Research and Development (R&D)**通过严格的研发流程来开发和优化产品。
   - **Objective**: Develop and optimize the product through rigorous R&D processes.
6. **Prototyping and Testing**创建原型并进行全面的测试，以确保其有效性和安全性。
   - **Objective**: Create prototypes and conduct comprehensive testing to ensure efficacy and safety.
7. **Regulatory Approval**获得对该产品的必要的监管批准。
   - **Objective**: Obtain necessary regulatory approvals for the product.
8. **Production Planning**：计划和建立生产流程和设施。
   - **Objective**: Plan and establish production processes and facilities.
9. **Market Analysis and Strategy**了解市场需求，制定市场进入策略。
   - **Objective**: Understand market needs and develop a go-to-market strategy.
10. **Launch and Post-Launch Evaluation**推出该产品，并持续监控其在市场上的表现。
    - **Objective**: Launch the product and continuously monitor its performance in the market.

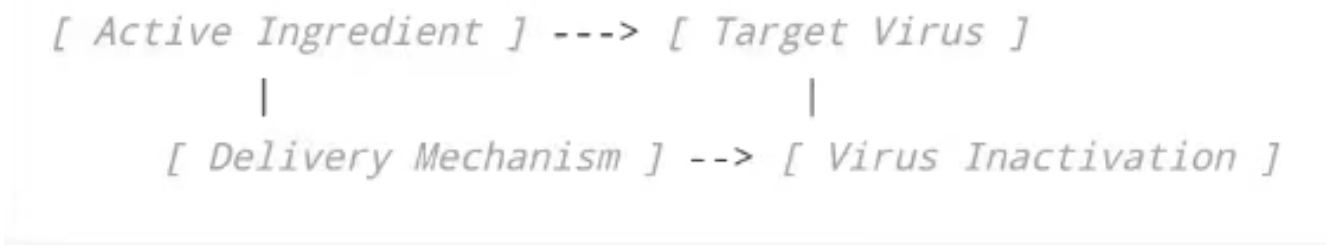
3) True Needs of the Product and Key Specifications#### 3)产品的真实需求和关键规格

The new product is a chemical medicine aimed at treating COVID-19 or preventing its spread. The true needs and key specifications include:产品的真实需求和关键规格 新产品是一种旨在治疗或防止其传播的化学药物。真正的需求和关键规格包括：

- **Efficacy**: High effectiveness against SARS-CoV-2 virus.对 SARS-CoV-2 病毒具有高效。
- **Safety**: Minimal side effects and high safety profile.副作用最小，安全性较高
- **Ease of Use**: Suitable for oral or intravenous administration.适合口服或静脉注射。
- **Stability**: Long shelf life under various storage conditions.在不同储存条件下的保质期长。
- **Affordability**: Cost-effective for mass production and distribution.对生产和分销具有成本效益。
- **Scalability**: Easily scalable manufacturing process.易于扩展的制造过程。

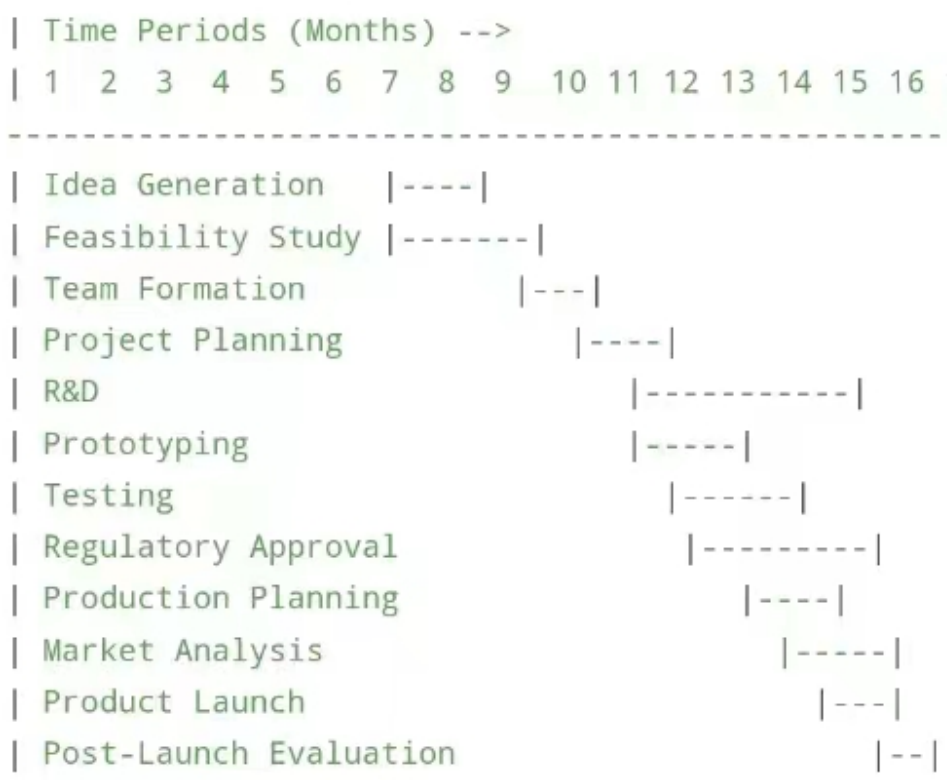
4) Diagram Sketch4)图草图 下面是一种抗病毒化学药物的简单概念图：

Below is a simple conceptual diagram of an antiviral chemical drug:



5) Missing Information to be Collected5)缺少要收集的信息

- Detailed molecular properties of the active ingredient.-活性成分的详细分子性质。
- Comprehensive pharmacokinetics and pharmacodynamics data.-全面的药代动力学和药效学数据。
- Clinical trial data to ensure safety and efficacy.-临床试验数据，以确保安全性和有效性。
- Regulatory guidelines specific to COVID-19 treatments.-专门针对 COVID-19 治疗的监管指南
- Market analysis reports on similar products.-关于类似产品的市场分析报告。
- Supply chain logistics for raw materials and distribution.-原材料和分销的供应链物流。



Investment Plan:

- Initial Research and Feasibility: $50,000
- Team Formation and Salaries: $100,000
- R&D and Prototyping: $200,000
- Testing and Regulatory Compliance: $150,000
- Production Setup: $300,000
- Marketing and Launch: $100,000
- Contingency Fund: $100,000
- **Total Estimated Investment**: $1,000,000

填空题

The chemical product in the Chemical Product Engineering refers to the functional products that are related to chemicals, chemical reactions or principles, and are used directly by the end consumers to satisfy their targeted purposes. The product may be a real or virtual one.化工产品工程中的化学产品，是指与化学产品、化学反应或原理有关，由最终消费者直接使用以满足其目标目的的功能性产品。该产品可以是真实的产品，也可以是虚拟的产品。

Examples of virtual chemical products include **(a) **chemical simulation software** and (b) **process modeling tools**.** 虚拟化学产品的例子包括(a)**化学模拟软件**和(b)**过程建模工具**。

The real chemical products vary greatly from specialty chemicals that are chemicals with special functionality like **(c) **photoinitiators** and (d) **corrosion inhibitors**,** 真正的化学产品与具有特殊功能的特殊化学品差别很大，如(c)**光引发剂**和(d)**腐蚀抑制剂**，

to formulated products that are composite with different components and functionality like **(e) **paints** and (f) **cosmetics**,** 对于具有不同成分和功能的复合配方产品，如(e)油漆和(f)化妆品，

to devices that are related to specific chemical reactions/principles, like (g) ****fuel cells** and (h) **catalytic converters**,** 对于与特定化学反应/原理有关的装置，如(g)燃料电池和(h)催化转换器，

to bio-based chemical products, like (i) ****biofuels** and (j) **biodegradable plastics**;** 到生物基化学产品，如(i)生物燃料和(j)可生物降解塑料；

and even to micro-structured products for which the micro-sized structure dominates their property and functionality like (k) ****drug delivery systems** and (l) **microfluidic devices**.**甚至是由微结构主导其性能和功能的微结构产品，如(k)药物传递系统和(l)微流控设备。

For the micro-structured products, their properties not only depend on **(m) ** composition **** that controls their thermodynamic property, but also **on (n) **structure**** that controls their dynamic property. 对于微结构产品，它们的特性不仅取决于控制其热力学特性的(m)组成，还取决于控制其动态特性的(n)结构。

On the contrary, the bulk or commodity chemicals that are usually produced hundreds of thousands tones annually do not belong to the scope of the chemical product design like (**o) **ethylene** and (p) **ammonia**.**相反，通常每年生产数十万吨的散装或商品化学品不属于(o)乙烯和(p)氨等化学产品设计的范围。